Experiment 1

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1. **AIM:** To design and implement the foundational frontend architecture of the EcoTrack application using modern React practices, Vite tooling, and ES6+ JavaScript features.
2. **Objective:**
   a. To set up a React project using Vite with proper project structure
   b. To understand component-based architecture in React
   c. To apply ES6 array methods (map, filter, reduce) for data-driven UI rendering
   d. To separate concerns using components, pages, and data modules
3. **Implementation/Code:**
   a. **Dashboard.jsx**

```
import logs from "../data/logs";



const Dashboard = ()=>{
    const sum = logs.reduce((acc, ele) => {
        return acc + ele.carbon;
    }, 0);

    return(
        <div style={{marginLeft: 'auto', backgroundColor:
"#323232ff",marginRight: "auto", maxWidth: "30%", padding: 20,
borderRadius: 10, borderColor: "blue", borderWidth: 2,
```

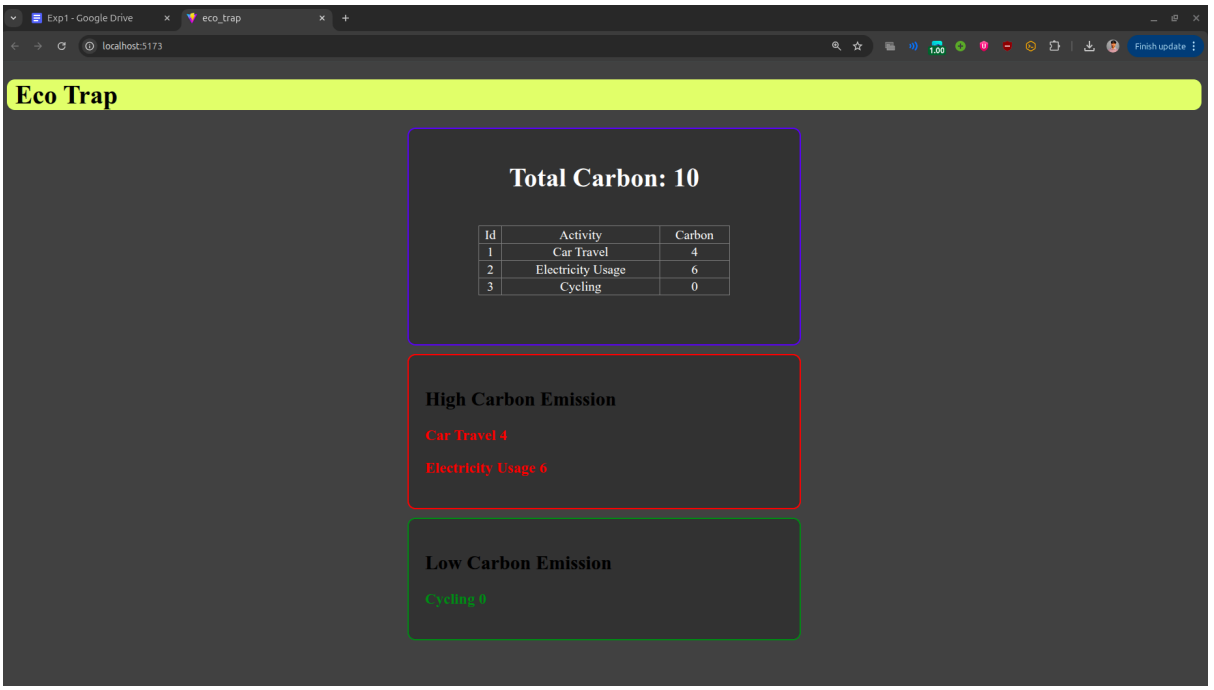
```
borderStyle: "solid"}}>
            <h1 style={{textAlign: 'center', color:
"#ffff"}}>Total Carbon: {sum}</h1>
            <table style={{
            margin: "40px auto",
            borderCollapse: "collapse",
            color: "#fff",
            width: "70%",
            textAlign: "center"
            }}
            border="1">
                <tr>
                    <td>Id</td>
                    <td>Activity</td>
                    <td>Carbon</td>
                </tr>
                {logs.map((element, index)=>{
                    return (
                        <tr>
                            <td>{element.id}</td>
                            <td>{element.activity}</td>
                            <td>{element.carbon}</td>
                        </tr>
                    )
                })}
            </table>
        </div>
    );
};
export default Dashboard;
```

b. Logs.jsx

```
import logs from "../data/logs";

const Logs = ()=>{
    return(
        <>
        <div style={{marginLeft: 'auto', backgroundColor:
"#323232ff",marginRight: "auto", maxWidth: "30%", padding:
20,marginBottom: 10, marginTop: 10, borderRadius: 10,
```

```
borderColor: "red", borderWidth: 2, borderStyle: "solid"}}>
            <h2>High Carbon Emission</h2>
            {logs.filter((ele)=>{
                return ele.carbon >= 4;
            }).map((ele)=> {
                return(
                    <>
                        <h3 style={{color: "red"}}>{ele.activity}
{ele.carbon}</h3>
                    </>
                )
            })}
        </div>

        <div style={{marginLeft: 'auto', backgroundColor:
"#323232ff",marginRight: "auto", maxWidth: "30%", padding: 20,
borderRadius: 10, borderColor: "green", borderWidth: 2,
borderStyle: "solid"}}>
            <h2>Low Carbon Emission</h2>

            {logs.filter((ele)=>{
                return ele.carbon < 4;
            }).map((ele)=> {
                return(
                    <>
                        <h3 style={{color:
"Green"}}>{ele.activity} {ele.carbon}</h3>
                    </>
                )
            })}
        </div>

        </>

    )
}
export default Logs;
```

4. Output:



5. Learning Outcome:

a. I learnt how to make react applications.
b. How the react dom actually works.
c. I learnt how to use the filter, map and the reduce function.